

# Clinical Profiles of Advanced Therapeutic Peptides, Incretin Mimetics, and Bioactive Compounds

## GLP-3 Retatrutide (Triple Receptor Agonist)

### Mechanism of Action

GLP-3 Retatrutide is an investigational, single-peptide unimolecular compound designed to simultaneously target and activate three distinct G-protein-coupled receptors key to human metabolic regulation: the glucagon-like peptide-1 (GLP-1) receptor, the glucose-dependent insulinotropic polypeptide (GIP) receptor, and the glucagon receptor.<sup>1</sup> This triple agonism represents a major pharmacological advancement over traditional mono-agonists such as semaglutide and dual-agonists such as tirzepatide.<sup>1</sup>

Upon administration, the GLP-1 and GIP receptor pathways work in concert to stimulate insulin secretion from pancreatic  $\beta$ -cells in a glucose-dependent manner, delay gastric emptying, and act on hypothalamic POMC/CART neural pathways to suppress appetite and increase early satiation.<sup>1</sup>

The integration of glucagon receptor activation introduces a complementary metabolic mechanism.<sup>1</sup> While GLP-1 and GIP agonism primarily focus on calorie restriction and insulin modulation, the glucagon pathway directly increases energy expenditure, activates hepatic mitochondrial activity, and accelerates lipid oxidation.<sup>1</sup> This multi-pathway activation addresses both energy intake and metabolic output, promoting fat-mass reduction while preserving systemic glycemic stability.<sup>1</sup>

### Therapeutic Benefits

GLP-3 Retatrutide addresses severe obesity, type 2 diabetes mellitus, metabolic dysfunction-associated steatotic liver disease (MASLD), and elevated cardiovascular risk.<sup>2</sup> Its primary therapeutic benefit lies in its ability to induce weight loss at a magnitude historically achieved only through bariatric surgery, helping resolve clinical obesity and reduce visceral adiposity.<sup>2</sup>

For patients with type 2 diabetes, retatrutide optimizes glycemic control, driving HbA1c levels toward normoglycemia.<sup>4</sup> Additionally, it improves lipid markers and lowers systemic inflammation, reducing cardiovascular risk factors such as waist circumference, non-HDL cholesterol, triglycerides, and systolic blood pressure.<sup>2</sup>

## Supporting Clinical Evidence

The clinical efficacy of retatrutide is established through Phase 2 and Phase 3 trial data.<sup>2</sup> In the Phase 3 TRIUMPH-1 trial, which evaluated adults with obesity or overweight without diabetes, weekly subcutaneous injections of retatrutide yielded significant weight reductions over 80 weeks.<sup>2</sup>

Participants on the highest dose of 12 mg achieved an average body weight loss of 24.2% (26.2 kg), while those on the lowest 1 mg dose lost 13.5% (12.9 kg), compared to a 1.6% reduction in the placebo group (1.5 kg).<sup>2</sup> Notably, 26% of the 12 mg cohort achieved  $\geq 30\%$  weight loss.<sup>2</sup> Additionally, 100% of participants on the highest dose achieved a BMI of less than 30, resolving their clinical classification of obesity.<sup>2</sup>

In a 104-week extension study of 532 participants with a baseline BMI  $\geq 35$ , those maintaining the 12 mg dose reached a mean weight loss of 30.3% (38.5 kg).<sup>2</sup> This weight loss was accompanied by significant improvements in blood pressure, fasting insulin, and circulating lipids.<sup>2</sup>

For diabetic populations, the Phase 3 TRANSCEND-T2D-1 trial demonstrated that once-weekly retatrutide monotherapy over 40 weeks significantly improved glycated hemoglobin concentrations.<sup>4</sup> The 12 mg dose cohort experienced an average HbA1c reduction of 2.1% from baseline, enabling 72% to 82% of patients to reach normoglycemic HbA1c levels ( $<5.7\%$ ) without severe hypoglycemic events.<sup>4</sup>

The primary tolerability concerns were gastrointestinal symptoms (nausea, diarrhea, constipation, and vomiting).<sup>2</sup> These events were mild-to-moderate and occurred primarily during dose escalation.<sup>2</sup> Mild dysesthesia was also reported in up to 12.5% of patients in the 12 mg group.<sup>2</sup>

| Clinical Trial Parameter                      | Placebo Cohort | Retatrutide 4 mg | Retatrutide 9 mg | Retatrutide 12 mg |
|-----------------------------------------------|----------------|------------------|------------------|-------------------|
| 80-Week Weight Loss (TRIUMPH-1) <sup>2</sup>  | -2.2%          | -19.0%           | N/A              | 24.2%             |
| 104-Week Weight Loss (TRIUMPH-1) <sup>2</sup> | N/A            | -27.9%           | -29.5%           | -30.3%            |

|                                                                     |        |        |        |                          |
|---------------------------------------------------------------------|--------|--------|--------|--------------------------|
| <b>40-Week HbA1c Reduction (TRANSCEND-T2D-1) <sup>4</sup></b>       | −0.81% | −1.69% | −1.86% | −1.94%                   |
| <b>Resolution of Obesity (BMI &lt; 30 at 80 weeks) <sup>2</sup></b> | Low    | N/A    | N/A    | <b>65.3%</b> of patients |
| <b>Rate of Nausea in TRIUMPH-1 <sup>2</sup></b>                     | 14.8%  | 28.6%  | 38.4%  | 42.4%                    |
| <b>Rate of Vomiting in TRIUMPH-1 <sup>2</sup></b>                   | 4.8%   | 10.6%  | 22.8%  | 25.3%                    |

## GHK-Cu (Copper Tripeptide-1)

### Mechanism of Action

GHK-Cu is a naturally occurring copper-binding tripeptide (Glycyl-L-Histidyl-L-Lysine) widely distributed in human plasma, saliva, and urine.<sup>6</sup> The peptide exhibits high affinity for copper ( $2+$ ) ions, forming the chelate GHK-Cu.<sup>6</sup> In this complex, the divalent metal ion is coordinated by specific nitrogens on its histidyl imidazole group, glycine amine group, and deprotonated peptide backbones.<sup>6</sup>

This coordination shields the redox activity of copper, preventing the generation of free radicals while facilitating the delivery of non-toxic, bioavailable copper to intracellular metalloenzymes.<sup>6</sup> These include lysyl oxidase (essential for collagen cross-linking), superoxide dismutase (a key antioxidant), and cytochrome c oxidase (involved in cellular respiration).<sup>6</sup>

GHK-Cu is released locally during tissue injury via the proteolytic degradation of extracellular matrix proteins, specifically SPARC and type I collagen.<sup>6</sup> As a "matrikine," it coordinates tissue repair by acting as a chemoattractant for macrophages, mast cells, and capillary endothelial cells, directing them to the wound site.<sup>6</sup>

At the genomic level, GHK-Cu up- and downregulates approximately 4,000 human genes, shifting the cellular transcriptome toward a healthy, regenerative state.<sup>6</sup> It stimulates the expression of key structural proteins, including collagen and elastin, and upregulates glycosaminoglycans, chondroitin sulfate, and the small leucine-rich proteoglycan decorin.<sup>6</sup>

GHK-Cu also regulates the activity of matrix metalloproteinases (MMPs) and their tissue inhibitors (TIMP-1 and TIMP-2), maintaining a balanced microenvironment for tissue remodeling.<sup>6</sup>

## Therapeutic Benefits

GHK-Cu addresses impaired wound healing (particularly in ischemic, diabetic, and elderly tissues), radiation-induced dermal damage, chronic obstructive pulmonary disease (COPD), cellular senescence, and age-related skin deterioration.<sup>6</sup>

In wound care, GHK-Cu accelerates wound closure and contraction, reduces local inflammation, and promotes healthy neovascularization.<sup>6</sup> Its anti-aging benefits include improving skin density, elasticity, and firmness, while reducing fine lines, wrinkles, and hyperpigmentation.<sup>6</sup>

Furthermore, GHK-Cu holds potential therapeutic value in oncology and pulmonology, where its ability to modulate metastatic gene networks and reverse pathological emphysematous tissue signatures supports organ-level recovery.<sup>6</sup>

## Supporting Clinical Evidence

Gene profiling research using Broad Institute databases has shown that GHK-Cu downregulates metastatic pathways.<sup>6</sup> In a study analyzing aggressive colorectal cancer signatures, GHK-Cu at nanomolar concentrations suppressed the expression of **70%** of 54 genes associated with metastasis.<sup>6</sup> This included downregulating node molecules such as APP, LMNA, and YWHAB.<sup>6</sup>

For tissue remodeling, GHK-Cu has been studied as a prospective therapeutic agent for COPD. It reversed emphysematous gene expression signatures in lung tissue by activating the TGF- $\beta$  pathway and elevating integrin  $\beta$ 1 expression in lung fibroblasts, restoring their ability to contract and remodel collagen.<sup>6</sup>

In stem cell research, GHK-Cu ( **0.1  $\mu$ M** to **10  $\mu$ M** ) increased the expression of epidermal stem cell markers, such as integrins and transcription factor p63, in basal keratinocytes, restoring their proliferative capacity.<sup>6</sup>

In animal models, GHK-Cu accelerated healing across diabetic and ischemic wounds, decreased the pro-inflammatory cytokine TNF- $\alpha$ , and elevated antioxidant enzyme levels in rats, mice, pigs, and dogs.<sup>6</sup>

Controlled clinical studies in human dermatology have validated these regenerative properties.<sup>6</sup> In placebo-controlled, split-face trials, GHK-Cu eye and facial creams stimulated collagen production, significantly reduced the appearance of fine lines and coarse wrinkles, and improved skin clarity, density, and elasticity.<sup>6</sup>

| Biological Target / Pathway                                                           | Observed Molecular and Cellular Response                                           | Clinical Outcome                                           |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Genomic Reset (Broad Institute)</b> <sup>6</sup>                                   | Modulated expression of $\approx$ genes back to healthier states                   | Broad systemic rejuvenation and cellular defense           |
| <b>TGF-<math>\beta</math> / Integrin <math>\beta_1</math> Activation</b> <sup>6</sup> | Reversed emphysematous gene signatures in lung fibroblasts                         | Pulmonary tissue repair and potential COPD mitigation      |
| <b>Oncological Node Suppression</b> <sup>6</sup>                                      | Suppressed <b>70%</b> of 54 metastatic colorectal cancer genes                     | Downregulated aggressive tumor pathways (e.g., YWHAB, APP) |
| <b>Epidermal Stem Cells (Keratinocytes)</b> <sup>6</sup>                              | Upregulated integrins and p63 stemness markers ( <b>0.1–10 <math>\mu</math>M</b> ) | Restored proliferative potential of senescent skin layers  |
| <b>Extracellular Matrix (ECM) Synthesis</b> <sup>6</sup>                              | Stimulated collagen, elastin, dermatan sulfate, and decorin                        | Enhanced skin density, elasticity, and accelerated healing |

## Wolverine (BPC-157 / TB-500 Stack)

### Mechanism of Action

The "Wolverine" stack is a synergistic peptide combination consisting of BPC-157 (Body Protection Compound-157) and TB-500 (a synthetic analog of the naturally occurring peptide Thymosin Beta-4).<sup>9</sup> These peptides target overlapping but distinct biological repair pathways, creating a synergistic effect that accelerates tissue regeneration.<sup>11</sup>

BPC-157 is a 15-amino-acid synthetic peptide derived from a protective protein found in human gastric juice.<sup>9</sup> It acts locally at the site of injury to accelerate neovascularization by upregulating vascular endothelial growth factor (VEGF) expression through the FAK/AKT/eNOS signaling cascade.<sup>13</sup> It modulates nitric oxide (NO) synthesis, which improves microvascular blood flow and reduces localized inflammatory cascades.<sup>11</sup> BPC-157 also stimulates fibroblast proliferation and activates growth hormone receptors in tendon fibroblasts, promoting the deposition of well-organized type I collagen.<sup>10</sup>

TB-500 is a synthetic version of Thymosin Beta-4, a 43-amino-acid peptide present in almost

all human cells.<sup>10</sup> Unlike the localized action of BPC-157, TB-500 operates systemically to coordinate early-stage wound healing.<sup>10</sup> Its primary mechanism is the regulation of monomeric G-actin.<sup>10</sup>

By sequestering G-actin, TB-500 regulates actin polymerization and cytoskeletal remodeling, which are essential for cellular motility, migration, and tissue repair.<sup>10</sup> TB-500 also downregulates pro-inflammatory cytokines, suppresses myofibroblast differentiation to minimize fibrotic scarring, and mobilizes stem and progenitor cells to damaged tissues.<sup>11</sup>

Therapeutic Benefits

The Wolverine stack addresses acute and chronic musculoskeletal injuries, including tendon ruptures, ligament tears, muscle strains, joint wear-and-tear, post-surgical recovery, and impaired wound healing.<sup>10</sup> It also supports mucosal cytoprotection, assisting in the repair of intestinal permeability and leaky gut.<sup>10</sup>

Combining these peptides provides dual-layered healing: BPC-157 drives focal regeneration, collagen deposition, and localized vascular supply, while TB-500 systemically mobilizes progenitor cells and regulates cell migration to the injury site.<sup>10</sup> This coordinated action accelerates recovery, reduces downstream pain, and improves the mechanical and functional strength of the repaired tissue.<sup>10</sup>

Supporting Clinical Evidence

The therapeutic synergy of the Wolverine stack is observed across three distinct physiological phases:

In preclinical animal models, this dual action has been shown to improve functional, structural, and biomechanical outcomes in skeletal muscle, tendon, and ligament injury models.<sup>15</sup> Chronic BPC-157 exposure consistently promoted bone-to-tendon junction repair and healed complex musculoskeletal lesions when administered either systemically or locally.<sup>15</sup>

Clinical data on the individual peptides and the stack continue to grow.<sup>16</sup> In a 2024 human pilot study evaluating patients with chronic, non-responsive knee pain, intra-articular administration

of BPC-157 yielded long-term relief ( > 6 months) in 7 out of 12 patients.<sup>16</sup>

Concurrently, Thymosin Beta-4 has demonstrated clinical efficacy in Phase II randomized clinical trials, showing accelerated corneal wound repair (via the RGN-259 clinical protocol) and promoting dermal repair in patients with pressure ulcers, venous stasis ulcers, and epidermolysis bullosa.<sup>17</sup> Intravenous safety evaluations in 2025 confirmed that BPC-157 was well-tolerated in healthy adults, showing no adverse events across vital organ systems.<sup>16</sup>

|                       |                                           |                                    |                            |
|-----------------------|-------------------------------------------|------------------------------------|----------------------------|
| Recovery<br>Parameter | BPC-157 (Gastric<br>Pentadecapeptide<br>) | TB-500 (Thymosin<br>Beta-4 Analog) | Wolverine Stack<br>Synergy |
|-----------------------|-------------------------------------------|------------------------------------|----------------------------|

|                               |                                                                 |                                                                      |                                                                                    |
|-------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Primary Site of Action</b> | Localized tissue repair and mucosal lining <sup>11</sup>        | Systemic cellular migration and motility <sup>11</sup>               | Combined local neovascularization and systemic stem cell recruitment <sup>11</sup> |
| <b>Angiogenesis Pathway</b>   | Upregulates localized VEGF and nitric oxide (NO) <sup>11</sup>  | Stimulates blood vessel sprouting via actin dynamics <sup>10</sup>   | Compounded vascular supply to hypoxic, damaged tissues <sup>11</sup>               |
| <b>Collagen Regulation</b>    | Drives fibroblast proliferation & type I collagen <sup>10</sup> | Enhances collagen matrix flexibility & limits fibrosis <sup>12</sup> | Upregulates collagen deposition while preventing rigid scar tissue <sup>11</sup>   |
| <b>Inflammatory Profile</b>   | Modulates local oxidative stress and cytokines <sup>12</sup>    | Downregulates systemic pro-inflammatory signaling <sup>11</sup>      | Manages the inflammatory cascade, reducing acute swelling <sup>11</sup>            |

## Semax

### Mechanism of Action

Semax is a synthetic heptapeptide with the amino acid sequence Met-Glu-His-Phe-Pro-Gly-Pro.<sup>20</sup> It is a hybrid molecule designed by conjugating a fragment of adrenocorticotrophic hormone, ACTH(4–7), to the C-terminal tripeptide Pro-Gly-Pro (PGP).<sup>20</sup> This PGP conjugation enhances metabolic stability, rendering Semax resistant to rapid enzymatic degradation by blood peptidases, while avoiding the hormonal or adrenal activity of full-length ACTH.<sup>20</sup>

Semax readily crosses the blood-brain barrier.<sup>22</sup> It exerts its nootropic and neuroprotective effects through several molecular mechanisms:

1. **Neurotrophic Support:** Semax upregulates brain-derived neurotrophic factor (BDNF) and its high-affinity receptor, tyrosine kinase receptor B (TrkB), in the hippocampus.<sup>23</sup> This pathway is key to neurogenesis, synaptic plasticity, and long-term memory formation.<sup>23</sup>
2. **Neurotransmitter Modulation:** It modulates the central dopaminergic and serotonergic systems, raising extracellular levels and accelerating the turnover of both dopamine and serotonin.<sup>23</sup> This action supports executive function, motivation, and emotional regulation.<sup>23</sup>
3. **Genomic Transcriptome Alteration:** In response to ischemia, Semax alters the brain transcriptome, downregulating pro-inflammatory JNK signaling (pJNK), upregulating the

survival-related transcription factor pCREB, and decreasing the expression of matrix metalloproteinase-9 (MMP-9).<sup>22</sup>

4. **Anti-inflammatory & Calcium Regulation:** Semax dampens neuroinflammation by reducing the expression of major histocompatibility complex (MHC) class I and II genes, mitigating microglial overactivation.<sup>20</sup> It also preserves mitochondrial membrane stability and prevents intracellular calcium (Ca<sup>2+</sup>) toxicity under hypoxic conditions.<sup>20</sup>

## Therapeutic Benefits

Semax addresses acute ischemic stroke recovery, transient ischemic attacks, cognitive deficits associated with vascular dementia and Alzheimer's disease, traumatic brain injury (TBI), optic nerve atrophy, and severe neuroinflammatory states.<sup>22</sup>

Its primary therapeutic benefits include preserving neural tissue during hypoxic insults, accelerating functional rehabilitation post-stroke, and improving memory retention, focus, and overall learning capacity without the peripheral sympathetic activation common to central nervous system stimulants.<sup>22</sup>

## Supporting Clinical Evidence

Semax is approved in Russia and Ukraine for neurological indications, including acute ischemic stroke and vascular cognitive impairment.<sup>22</sup>

In rodent models of focal cerebral ischemia induced by permanent middle cerebral artery occlusion (pMCAO), genome-wide transcriptome profiling demonstrated that Semax significantly alters cortical tissue gene expression.<sup>20</sup>

- **3 Hours Post-Ischemia:** Semax altered the expression of 96 genes, upregulating key transcription regulators and modulating 24 genes involved in vascular endothelial migration, hematopoiesis, and vasculogenesis.<sup>20</sup> This early intervention promoted compensatory neovascularization around the ischemic core.<sup>20</sup>
- **24 Hours Post-Ischemia:** Semax altered the expression of 68 genes, with over 50% of them directly regulating the immune response.<sup>20</sup> This late-stage intervention upregulates chemokine and immunoglobulin genes, helping to balance post-ischemic neuroinflammation.<sup>20</sup>

Additionally, in transgenic APP<sup>swe</sup>/PS1<sup>dE9</sup> mouse models of Alzheimer's disease, intranasal Semax administration (1.25 mg/kg every other day for one month, starting at 6 months of age) significantly improved behavioral outcomes.<sup>21</sup> Treated mice showed improved cognitive function, spatial learning, and memory retention in Barnes maze and novel object recognition tests.<sup>21</sup> Histological analysis confirmed that Semax treatment led to a significant reduction in amyloid- $\beta$  plaques within both the cortex and the hippocampus.<sup>21</sup>

## Melanotan 2



## Mechanism of Action

Melanotan II (MT-II) is a synthetic, cyclic lactam heptapeptide analog of the endogenous hormone alpha-melanocyte-stimulating hormone ( $\alpha$ -MSH).<sup>25</sup> Its primary mechanism of action is non-selective agonism of the G-protein-coupled melanocortin receptors (MCRs), displaying high affinity for MC1, MC3, and MC4 receptors.<sup>25</sup>

- **MC1R Engagement:** MT-II binds to MC1R expressed on dermal melanocytes, stimulating melanogenesis and skin pigmentation independently of ultraviolet radiation.<sup>25</sup>
- **MC3R & MC4R Engagement:** It activates central MC3 and MC4 receptors in the hypothalamus, which are key regulators of energy balance, appetite, and sexual function.<sup>25</sup> MC4R activation in the paraventricular nucleus initiates erections through central neural pathways rather than peripheral vasodilation.<sup>25</sup> This pathway involves central oxytocinergic and dopaminergic projections to the spinal cord, driving erectile activity even in the absence of sexual stimulation.<sup>25</sup> Additionally, hypothalamic MC4R activation suppresses appetite and reduces binge-like ethanol consumption by modulating central reward circuitry.<sup>25</sup>

## Therapeutic Benefits

Melanotan II addresses psychogenic and organic erectile dysfunction (particularly in patients who are non-responsive to or cannot tolerate phosphodiesterase-5 inhibitors), hypoactive sexual desire disorder, pathologic obesity, and severe sun-induced dermal damage in fair-skinned phenotypes.<sup>25</sup>

Its primary benefit lies in its dual central action: it stimulates penile rigidity and elevates subjective sexual desire while also suppressing appetite, supporting weight management, and promoting protective skin pigmentation.<sup>25</sup>

## Supporting Clinical Evidence

The pro-erectile effects of Melanotan II have been evaluated in double-blind, placebo-controlled crossover clinical trials led by researchers such as H. Wessells.<sup>30</sup>

In an initial study of 10 men with psychogenic erectile dysfunction, a single subcutaneous dose of 0.025 mg/kg induced clinically apparent penile erections in 8 of 10 participants.<sup>30</sup> The mean duration of tip rigidity exceeding 80% was 38 minutes, compared to only 3 minutes in the placebo group.<sup>30</sup>

A follow-up study of 20 men with organic erectile dysfunction risk factors confirmed these findings, showing that MT-II administration produced robust erections in 17 participants, with high-rigidity episodes averaging 41 minutes.<sup>30</sup> Subjective sexual desire scores increased in 68% of active treatment sessions, compared to 19% with placebo.<sup>30</sup>

In preclinical studies, the necessity of the MC4R pathway was confirmed using Mc4r knockout rat models.<sup>27</sup> Agonist administration did not alter feeding or grooming behaviors in

homozygous knockouts, while wild-type controls exhibited characteristic appetite suppression and grooming responses.<sup>27</sup> Additionally, in mouse models of binge-like drinking, co-administering a low dose of MT-II ( **0.26 mg/kg** ) with naltrexone resulted in a synergistic 7.6-fold increase in the drug's effectiveness in reducing binge ethanol intake and lower blood alcohol concentrations.<sup>29</sup> Despite its efficacy, clinical development of MT-II was discontinued due to its side-effect profile, which included transient blood pressure elevation, flushing, frequent yawning, and severe nausea in approximately 50% of participants.<sup>28</sup>

| Parameter                                                 | Placebo Cohort | Melanotan II (0.025 mg/kg SubQ) | Clinical / Preclinical Context            |
|-----------------------------------------------------------|----------------|---------------------------------|-------------------------------------------|
| Erectile Success Rate (Psychogenic ED) <sup>30</sup>      | Minimal        | 80% of participants (8 of 10)   | Single-dose crossover human trial         |
| Mean Tip Rigidity (>80%) Duration <sup>30</sup>           | 3 minutes      | 38 minutes                      | Measured via computerized monitoring      |
| Mean Duration of High Rigidity (Organic ED) <sup>30</sup> | N/A            | 41 minutes                      | Cohort with vasculogenic & neural factors |
| Increase in Subjective Sexual Desire <sup>30</sup>        | 19% of doses   | 68% of doses                    | Crossover self-reporting scale            |
| Synergy with Naltrexone in Binge Drinking <sup>29</sup>   | Baseline       | 7.6-fold increase in efficacy   | Preclinical binge-like ethanol model      |

## CJC-1295 / Ipamorelin

### Mechanism of Action

The combination of CJC-1295 and Ipamorelin represents a dual-receptor growth hormone secretagogue (GHS) strategy designed to optimize the secretion of endogenous growth hormone (GH).<sup>32</sup>

CJC-1295 is a synthetic 29-amino-acid peptide analog of endogenous Growth Hormone-Releasing Hormone (GHRH).<sup>32</sup> Its modification with a Drug Affinity Complex (DAC) allows it to bind covalently to circulating albumin, extending its biological half-life from minutes to approximately 5.8–8.1 days.<sup>32</sup> CJC-1295 binds to GHRH receptors on pituitary somatotrophs, upregulating cAMP-dependent pathways to stimulate growth hormone synthesis and baseline release.<sup>33</sup>

Ipamorelin is a synthetic pentapeptide (Aib-His-D-2-Nal-D-Phe-Lys-NH<sub>2</sub>) that acts as a highly selective agonist of the growth hormone secretagogue receptor type 1a (GHS-R1a), also known as the ghrelin receptor.<sup>35</sup> It binds to GHS-R1a on somatotrophs, activating the Gαq-phospholipase C cascade.<sup>35</sup> This pathway triggers IP3-mediated calcium release from intracellular stores, stimulating the exocytotic release of stored GH in pulsatile waves.<sup>35</sup>

Ipamorelin is highly selective; unlike older secretagogues such as GHRP-2 or GHRP-6, its specific binding conformation does not stimulate corticotroph (ACTH-secreting) or lactotroph (prolactin-secreting) cells.<sup>35</sup> Consequently, it promotes robust GH release without elevating cortisol, prolactin, or ACTH.<sup>35</sup>

Together, CJC-1295 and Ipamorelin exert a synergistic effect.<sup>33</sup> While CJC-1295 increases the overall amplitude of GH release, Ipamorelin enhances pulse frequency, driving the hepatic production of insulin-like growth factor-1 (IGF-1) within physiological limits.<sup>33</sup>

## **Therapeutic Benefits**

The CJC-1295/Ipamorelin combination addresses age-related somatopause, adult growth hormone deficiency, muscle wasting, elevated visceral adiposity, poor sleep architecture, and impaired tissue recovery.<sup>32</sup>

By stimulating endogenous GH production, it supports protein synthesis, enhances nitrogen retention, and promotes lipolysis to optimize body composition.<sup>32</sup> It also accelerates tissue repair in muscles, tendons, and ligaments without the risk of pituitary desensitization or the supraphysiological spikes associated with exogenous human growth hormone (HGH) therapy.<sup>32</sup>

## **Supporting Clinical Evidence**

In a randomized, placebo-controlled, double-blind ascending-dose trial, healthy adults aged 21 to 61 received subcutaneous injections of CJC-1295. A single dose of 30 or 60 µg/kg produced a sustained, dose-dependent 2- to 10-fold increase in mean plasma GH concentrations for at least 6 days, and a 1.5- to 3-fold elevation in mean plasma IGF-1 concentrations lasting 9 to 11 days.<sup>32</sup>

Repeated administration demonstrated a cumulative effect, keeping IGF-1 levels elevated for up to 28 days post-treatment with no serious adverse reactions.<sup>34</sup>

Preclinical rodent models have further characterized Ipamorelin's efficacy.<sup>38</sup> In

hypophysectomized rats, Ipamorelin administration induced longitudinal bone growth, while studies in adult female rats showed a significant increase in bone mineral content.<sup>38</sup> Chronic exposure did not alter somatotroph cell morphology, though it did increase the volume density of secretagogue secretory granules.<sup>38</sup> Ipamorelin was also evaluated in a Phase II randomized controlled trial of 117 patients undergoing bowel resection to test its gastroprokinetic properties in postoperative ileus.<sup>38</sup> While the treatment was well-tolerated and showed a numerical trend toward faster recovery of bowel function, it did not meet its primary efficacy endpoints, leading to the discontinuation of its development for that specific indication.<sup>38</sup>

## KLOW

### Mechanism of Action

KLOW is a synthetic four-peptide compound formulated to coordinate tissue repair and modulate inflammatory pathways.<sup>39</sup> It contains GHK-Cu (50 mg), BPC-157 (10 mg), TB-500 (10 mg), and KPV (10 mg) in a single vial.<sup>41</sup> This combination targets multiple cellular pathways involved in tissue repair.<sup>40</sup>

- **GHK-Cu & BPC-157:** GHK-Cu upregulates cellular genes involved in wound healing and remodels the extracellular matrix, while BPC-157 stimulates localized neovascularization, fibroblast proliferation, and osteotendinous repair.<sup>6</sup>
- **TB-500:** TB-500 coordinates systemic cell migration and stem cell recruitment to prevent disorganized scarring.<sup>40</sup>
- **KPV:** KPV (Lys-Pro-Val) is a highly conserved tripeptide derived from the C-terminal sequence of  $\alpha$ -melanocyte-stimulating hormone ( $\alpha$ -MSH).<sup>44</sup> KPV is transported across cell membranes via Peptide Transporter 1 (PepT1), which is expressed on immune and intestinal epithelial cells.<sup>43</sup> Once inside the cell, KPV inhibits both NF- $\kappa$ B and MAP kinase inflammatory signaling pathways, reducing the transcription of pro-inflammatory cytokines such as TNF- $\alpha$ , IL-1 $\beta$ , and IL-6.<sup>43</sup> KPV also suppresses neutrophil migration and inhibits neutrophil extracellular trap (NET) formation (NETosis) by mitigating intracellular oxidative stress.<sup>46</sup>

### Therapeutic Benefits

KLOW addresses systemic inflammatory disorders, acute and chronic musculoskeletal trauma, impaired wound healing, inflammatory bowel diseases (such as ulcerative colitis and Crohn's disease), and skin aging.<sup>39</sup>

The quad-peptide synergy targets multiple aspects of recovery: the KPV and BPC-157 combination protects mucosal barriers, supports gut lining integrity, and modulates autoimmune inflammation.<sup>40</sup>

Concurrently, GHK-Cu and TB-500 promote cell migration to micro-injuries while preventing excessive scarring, accelerating recovery from physical stress, surgical procedures, and aesthetic treatments.<sup>39</sup>

## Supporting Clinical Evidence

While clinical data on the combined KLOW formulation are developing, peer-reviewed research on its individual components demonstrates robust therapeutic efficacy.<sup>40</sup>

In vitro, nanomolar concentrations of KPV inhibited NF-κB activation and reduced cytokine secretion in human intestinal epithelial and T-cells.<sup>45</sup> In animal models, oral KPV administration significantly reduced the severity of dextran sulfate sodium (DSS)- and trinitrobenzenesulfonic acid (TNBS)-induced colitis, as demonstrated by lower histological inflammation scores and reduced pro-inflammatory cytokine expression.<sup>45</sup>

Additionally, an engineered KPV conjugate (proKPV) achieved a 3.8-fold greater colonic accumulation in murine colitis models compared to free KPV.<sup>46</sup> This formulation suppressed key markers of neutrophil activation, including myeloperoxidase, neutrophil elastase, and citrullinated histone H3 (CitH3), demonstrating therapeutic efficacy at a 20-fold lower dose.<sup>46</sup> When combined with BPC-157's mucosal protection and the tissue remodeling properties of GHK-Cu and TB-500, KLOW offers a comprehensive approach to healing.<sup>40</sup> Observational data indicate that patients experience initial reductions in pain and inflammation within 2 to 3 weeks, with deeper tissue repair, improved joint mobility, and gastrointestinal support fully manifesting over 6 to 12 weeks.<sup>40</sup>

| KLOW Component | Standard Dose / Vial | Molecular Target                        | Clinical Relevance                                                  |
|----------------|----------------------|-----------------------------------------|---------------------------------------------------------------------|
| GHK-Cu         | 50 mg                | Extracellular Matrix & DNA <sup>6</sup> | Remodels collagen, restores skin stem cells <sup>6</sup>            |
| BPC-157        | 10 mg                | VEGF, FAK/AKT/eNOS <sup>13</sup>        | Promotes localized healing, tendon & joint repair <sup>43</sup>     |
| TB-500         | 10 mg                | Monomeric G-actin <sup>10</sup>         | Systemic cell migration, reduces fibrotic scar tissue <sup>11</sup> |
| KPV            | 10 mg                | PepT1, NF-κB, MAPK <sup>45</sup>        | Downregulates chronic gut & skin                                    |

|  |  |  |                            |
|--|--|--|----------------------------|
|  |  |  | inflammation <sup>43</sup> |
|--|--|--|----------------------------|

## GLOW

### Mechanism of Action

GLOW is a tri-peptide blend designed to support dermal regeneration and structural tissue repair.<sup>13</sup> It contains GHK-Cu (30-50 mg), BPC-157 (5-10 mg), and TB-500 (10 mg).<sup>13</sup> This combination utilizes overlapping cellular pathways to promote tissue remodeling<sup>13</sup>:

- **Dermal & Matrix Rebuilding:** GHK-Cu upregulates dermal genes involved in wound healing, stimulates collagen and elastin synthesis, and coordinates copper delivery to promote vascular growth and antioxidant defense.<sup>13</sup>
- **Localized Vascularization:** BPC-157 accelerates tissue healing through the FAK/AKT/eNOS pathway, upregulating local VEGF expression and modulating nitric oxide to restore microvascular blood flow.<sup>13</sup> It also organizes the deposition of collagen fibrils at the injury site.<sup>13</sup>
- **Systemic Cell Migration:** TB-500 regulates monomeric G-actin to facilitate cytoskeletal remodeling, promoting systemic cell migration, neovascularization, and stem cell mobilization to damaged areas.<sup>13</sup> This pathway helps minimize the formation of rigid, dysfunctional scar tissue.<sup>47</sup>

### Therapeutic Benefits

GLOW addresses skin aging, photo-damaged skin, post-procedure dermal recovery (following microneedling, laser resurfacing, or chemical peels), acute musculoskeletal trauma, and tendon or ligament tears.<sup>13</sup>

By combining GHK-Cu, BPC-157, and TB-500, GLOW supports cellular recovery from the inside out.<sup>47</sup> It stimulates collagen production to restore skin firmness and reduce wrinkles, while also accelerating recovery times for musculoskeletal injuries and minimizing post-procedural swelling and redness.<sup>13</sup>

### Supporting Clinical Evidence

Preclinical evaluations of the GLOW components highlight their roles in structural modeling and cellular migration.<sup>48</sup>

The tissue-remodeling properties of GHK-Cu are well-documented.<sup>6</sup> In placebo-controlled clinical trials, GHK-Cu creams significantly increased skin thickness, density, and elasticity, while reducing coarse and fine wrinkles.<sup>6</sup>

TB-500 (Thymosin Beta-4) has also demonstrated robust wound-healing efficacy.<sup>51</sup> In rodent full-thickness wound models, topical or systemic administration of Thymosin Beta-4 increased wound re-epithelialization by **42%** at 4 days and **61%** at 7 days, significantly enhancing

collagen deposition and neovascularization.<sup>51</sup>

Furthermore, a synthetic heptapeptide representing the actin-binding domain of Thymosin Beta-4 (LKKTETQ) promoted dermal repair in diabetic and senescent mice at rates comparable to the parent molecule.<sup>52</sup>

The addition of BPC-157's local cytoprotection further accelerates structural repair, making the GLOW formulation highly effective for both dermal and musculoskeletal regeneration.<sup>13</sup>

| Formulation                     | Active Components                                                                                                                         | Target Indications                                                                                                                                              | Primary Therapeutic Mechanisms                                                                                                                                                                                                                                                      |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KLOW Blend</b> <sup>39</sup> | <ul style="list-style-type: none"><li>- GHK-Cu (50 mg)</li><li>- BPC-157 (10 mg)</li><li>- TB-500 (10 mg)</li><li>- KPV (10 mg)</li></ul> | <ul style="list-style-type: none"><li>- Autoimmune disorders</li><li>- IBD / Gut issues</li><li>- Chronic joint pain</li><li>- Post-treatment healing</li></ul> | <ul style="list-style-type: none"><li>- Modulates NF-<math>\kappa</math>B and MAP kinase pathways via PepT1 <sup>45</sup></li><li>- Suppresses NETosis and local/systemic inflammation <sup>46</sup></li><li>- Coordinates local and systemic tissue repair <sup>43</sup></li></ul> |
| <b>GLOW Blend</b> <sup>13</sup> | <ul style="list-style-type: none"><li>- GHK-Cu (30-50 mg)</li><li>- BPC-157 (5-10 mg)</li><li>- TB-500 (10 mg)</li></ul>                  | <ul style="list-style-type: none"><li>- Aesthetic skin aging</li><li>- Post-procedure dermal healing</li><li>- Acute muscle/tendon tears</li></ul>              | <ul style="list-style-type: none"><li>- Stimulates dermal collagen and elastin synthesis <sup>13</sup></li><li>- Upregulates local VEGF-mediated neovascularization <sup>13</sup></li><li>- Promotes cell migration and actin cytoskeletal remodeling <sup>13</sup></li></ul>       |

# MOTS-c

## Mechanism of Action

MOTS-c (Mitochondrial Open Reading Frame of the 12S rRNA Type-C) is a naturally occurring 16-amino-acid mitochondrial-derived peptide (MDP) encoded by the mitochondrial genome.<sup>53</sup> It acts as an endocrine signal and a nuclear transcriptional regulator, coordinating metabolic homeostasis.<sup>54</sup>

- **AMPK & Akt Pathway Activation:** Under resting conditions, MOTS-c is localized within the mitochondria, but under metabolic stress, it translocates to the nucleus in an AMPK-dependent manner.<sup>53</sup> In the nucleus, MOTS-c binds to transcription factors regulated by antioxidant response elements (ARE), upregulating gene expression involved in stress resistance and metabolic defense.<sup>53</sup> In skeletal muscle, MOTS-c activates the 5'-monophosphate-activated protein kinase (AMPK) and Akt signaling pathways, upregulating glucose transporter-4 (GLUT4) expression and promoting insulin-dependent glucose clearance into skeletal muscle cells for glycolysis.<sup>53</sup>
- **Anti-inflammatory Pathway Regulation:** MOTS-c reduces systemic pro-inflammatory cytokines while upregulating anti-inflammatory pathways.<sup>55</sup> It suppresses the MAP kinase/c-Fos pathway, inhibits formalin-induced activation of extracellular signal-regulated kinase (ERK), JNK, and P38, and reduces chronic cartilage degradation by downregulating the Nrf2/TXNIP/NLRP3 inflammasome axis.<sup>55</sup>

## Therapeutic Benefits

MOTS-c addresses insulin resistance, type 2 diabetes mellitus, metabolic syndrome, obesity, age-related sarcopenia, physical endurance decline, osteoporosis, gestational diabetes mellitus (GDM), and degenerative joint diseases like osteoarthritis.<sup>54</sup>

By acting as an endogenous metabolic regulator, MOTS-c restores insulin sensitivity in skeletal muscle, prevents diet-induced weight gain without hepatotoxicity, protects pancreatic  $\beta$ -cells, and limits degenerative changes in articular cartilage and bone.<sup>55</sup>

## Supporting Clinical Evidence

MOTS-c expression is highly responsive to physical exercise.<sup>55</sup> Human trials demonstrate that high-intensity exercise stimulates its expression, generating an 11.9-fold increase in skeletal muscle MOTS-c levels and a 1.5- to 1.6-fold rise in circulating plasma levels.<sup>55</sup> However, endogenous MOTS-c levels decline with age; plasma concentrations are approximately 21% higher in young adults compared to middle-aged adults, and 29% higher compared to the elderly.<sup>55</sup>

In preclinical models, restoring MOTS-c levels has shown significant metabolic benefits.<sup>54</sup> Systemic administration of MOTS-c in aged mice reversed age-related insulin resistance and improved overall glucose tolerance.<sup>55</sup> In mouse models of high-fat diet-induced obesity,



MOTS-c therapy prevented weight gain and hyperinsulinemia without causing hepatic toxicity.<sup>55</sup>

Furthermore, in mouse models of gestational diabetes mellitus (GDM) induced by a high-fat diet and streptozotocin (STZ), daily MOTS-c administration during pregnancy successfully alleviated maternal hyperglycemia, improved insulin sensitivity and glucose tolerance, protected pancreatic  $\beta$ -cells from STZ-induced apoptosis, and significantly reduced GDM-induced fetal mortality and abnormal birth weights.<sup>57</sup>

In a murine model of osteoarthritis, exogenous MOTS-c improved mitochondrial dysfunction in chondrocytes, inhibited NLRP3 inflammasome-mediated pyroptosis, promoted extracellular matrix synthesis, and slowed the physical degeneration of articular cartilage.<sup>56</sup>

## NAD (Nicotinamide Adenine Dinucleotide)

### Mechanism of Action

Nicotinamide Adenine Dinucleotide (NAD<sup>+</sup>) is a classical coenzyme that facilitates hydrogen transfer in essential metabolic redox reactions, including glycolysis, the tricarboxylic acid (TCA) cycle, and oxidative phosphorylation.<sup>58</sup> Beyond its metabolic role, NAD<sup>+</sup> acts as an obligate cosubstrate for several classes of enzymes that regulate cellular health.<sup>58</sup>

1. **Sirtuins (SIRT1–7):** These are NAD<sup>+</sup>-dependent deacetylases and deacylases.<sup>60</sup> SIRT1, 6, and 7 utilize nuclear NAD<sup>+</sup> to deacetylate histones and transcription factors, regulating stress resistance, DNA repair, and circadian rhythms via CLOCK/BMAL-driven pathways.<sup>58</sup> SIRT3, 4, and 5 utilize mitochondrial NAD<sup>+</sup> to optimize oxidative phosphorylation, fatty acid oxidation, and antioxidant defense through SOD2 activation.<sup>58</sup>
2. **Poly-ADP-Ribose Polymerases (PARPs):** These enzymes utilize NAD<sup>+</sup> to facilitate DNA repair and maintain genomic integrity.<sup>58</sup>
3. **CD38/CD157 Ectoenzymes:** These glycohydrolases consume NAD<sup>+</sup> to generate cyclic ADP-ribose, a key intracellular calcium signaling molecule.<sup>58</sup>

During the normal aging process, tissue NAD<sup>+</sup> levels decline significantly.<sup>58</sup> This depletion is driven by the age-associated downregulation of nicotinamide phosphoribosyltransferase (NAMPT)—the rate-limiting enzyme in the NAD<sup>+</sup> salvage pathway—and the chronic activation of PARP1 (triggered by progressive DNA damage) and CD38, which continuously deplete the cellular NAD<sup>+</sup> pool.<sup>58</sup>

Elevating NAD<sup>+</sup> levels through precursors such as Nicotinamide Riboside (NR) or Nicotinamide Mononucleotide (NMN) restores sirtuin activity, reverses age-related mitochondrial decay, and improves nuclear-mitochondrial communication.<sup>58</sup>

## Therapeutic Benefits

**NAD<sup>+</sup>** restoration addresses age-related metabolic decline, vascular aging, peripheral artery disease (PAD), age-associated cognitive decline, neurodegenerative disorders (such as Alzheimer's, Parkinson's, and Huntington's diseases), cardiac hypertrophy, and skeletal muscle atrophy.<sup>58</sup>

By restoring cellular **NAD<sup>+</sup>** pools, this therapy reactivates mitochondrial complex I, upregulates sirtuin-mediated antioxidant pathways, enhances DNA repair, reduces oxidative stress, and improves cerebrovascular and endothelial function.<sup>58</sup>

## Supporting Clinical Evidence

Restoring **NAD<sup>+</sup>** levels has demonstrated clear therapeutic potential in both clinical pilot trials and preclinical aging models.<sup>58</sup>

In a pilot, 4-week open-label clinical trial (NCT06534944), older adults with peripheral artery disease (PAD)—a condition characterized by accelerated vascular aging and endothelial dysfunction—received oral Nicotinamide Riboside (NR) supplementation.<sup>62</sup>

NR therapy led to significant improvements in vascular and cognitive function.<sup>62</sup> Treated patients showed a positive trend in peripheral endothelial function and a significant increase in cerebrovascular response within the left dorsolateral prefrontal cortex.<sup>62</sup>

Cognitive testing also revealed enhancements across multiple domains.<sup>62</sup> Serum analysis from treated patients confirmed that NR supplementation reduced endothelial oxidative stress, increased nitric oxide production, and improved mitochondrial respiratory efficiency.<sup>62</sup>

Preclinical studies have further characterized these molecular mechanisms.<sup>58</sup> Knocking out or chemically inhibiting PARP1 or CD38 in mice elevated systemic **NAD<sup>+</sup>** levels and restored SIRT1 and SIRT3 activity, protecting against metabolic decline, complex I deficiencies, and cardiac ischemic damage.<sup>58</sup>

In rodent models of AD, administering **NAD<sup>+</sup>** precursors restored mitochondrial function, upregulated PGC-1 $\alpha$ , reduced toxic amyloid- $\beta$  accumulation, and protected against memory loss.<sup>58</sup> CD38 knockout or PARP1 inhibition systemically elevated **NAD<sup>+</sup>** levels and activated sirtuins, protecting against cardiovascular pathology.<sup>58</sup>

| Enzyme / Protein Family | Coenzyme / Cosubstrate Role of NAD+ | Biological Function | Pathological Aging Impact |
|-------------------------|-------------------------------------|---------------------|---------------------------|
|-------------------------|-------------------------------------|---------------------|---------------------------|

|                                            |                                                            |                                                                              |                                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Sirtuins (SIRT1–7)</b><br><sup>58</sup> | Obligate<br>Cosubstrate<br>(Cleaved at<br>glycosidic bond) | Histone<br>deacetylation,<br>mitochondrial<br>efficiency, circadian<br>clock | Decreased <b>NAD<sup>+</sup></b><br>availability impairs<br>sirtuin activity,<br>driving metabolic<br>and mitochondrial<br>decay |
| <b>PARPs (PARP1)</b> <sup>58</sup>         | Obligate<br>Cosubstrate<br>(ADP-ribose<br>transfer)        | Facilitates DNA<br>double-strand<br>break repair and<br>genomic integrity    | Chronic DNA<br>damage leads to<br>overactivation of<br>PARP1, depleting<br>the cellular <b>NAD<sup>+</sup></b><br>pool           |
| <b>CD38 / CD157</b> <sup>58</sup>          | Obligate<br>Cosubstrate<br>(Glycohydrolase<br>pathway)     | Generates cyclic<br>ADP-ribose to<br>regulate calcium<br>signaling           | Age-associated<br>upregulation of<br>CD38 accelerates<br>the systemic<br>depletion of<br><b>NAD<sup>+</sup></b>                  |
| <b>NAMPT</b> <sup>58</sup>                 | Rate-Limiting<br>Enzyme (Salvage<br>Pathway)               | Synthesizes NMN<br>from nicotinamide                                         | Chronic<br>inflammation and<br>oxidative stress<br>suppress NAMPT,<br>impairing <b>NAD<sup>+</sup></b><br>biosynthesis           |

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